

12 **(long)** Two concentric spherical surfaces have radii of  $R_1$  and  $R_2$ , and each uniformly carries a charge of  $-Q$  and  $+Q$ , as shown in Fig. 7(a), respectively.

(a) What is the electric potential at  $R_2$  with respect to Earth? [7]

(b) What is the electric potential at  $R_1$  with respect to Earth? [8]

(c) The two concentric spherical surfaces now uniformly carry a charge of  $+Q_1$  and  $+Q_2$  as shown in Fig. 7(b). What is the electric potential at  $R_2$  and  $R_1$  with respect to Earth? [15]

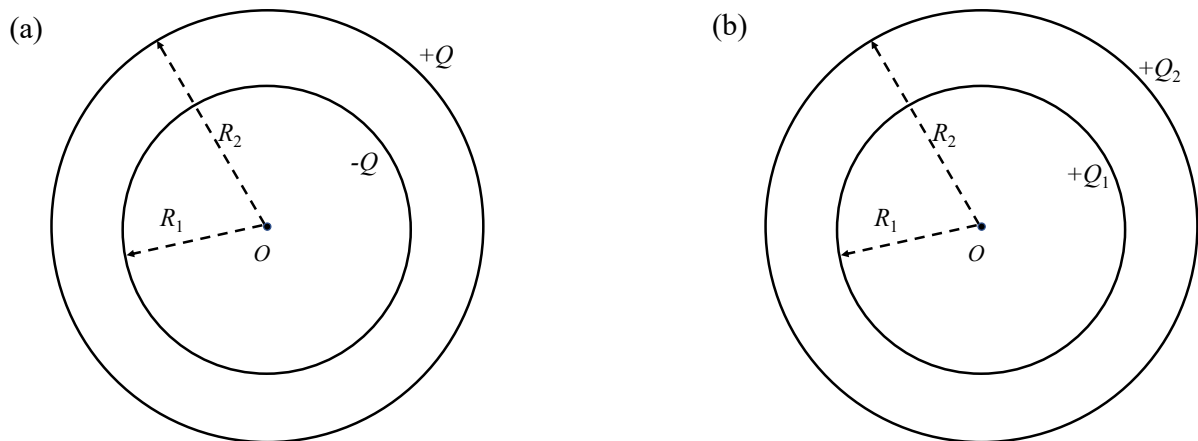


Fig. 7

Version OBA/2

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